

Procedural programming

Functional programming

Object Oriented Programming

- ✓

Video: Introduction to Object Oriented Programming
5 min
- ✓

Reading: OOP Principles
20 min
- ✓

Video: Python classes and instances
4 min
- ✓

Reading: Exercise: Define a Class
30 min
- ✓

Reading: Define a Class - solution
10 min
- 📖

Practice Quiz: Self-review: Define a Class
4 questions
- ✓

Video: Instantiate a custom Object
4 min
- ✓

Reading: Exercise: Instantiate a custom Object
30 min
- 📖

Reading: Instantiate a custom Object - solution
10 min
- 📖

Practice Quiz: Self-review: Instantiate a custom Object
4 questions
- ▶

Video: Instance methods
4 min
- ▶

Video: Parent classes vs. child classes
6 min
- 📖

Reading: Inheritance and Multiple Inheritance
30 min
- 📖

Reading: Exercise: Classes and object exploration
30 min
- ▶

Video: Abstract classes and methods
4 min
- 📖

Practice Quiz: Abstract classes and methods
5 questions
- 🔗

Programming Assignment: Abstract Classes and Methods
3h
- ▶

Video: Method Resolution Order
5 min
- 📖

Reading: Working with Methods: Examples
20 min
- 📖

Reading: Exercise: Working with Methods
30 min
- 📖

Reading: Working with Methods - solution
10 min
- 📖

Practice Quiz: Self-review: Working with Methods
6 questions
- ▶

Video: Module summary: Programming paradigms
2 min
- 📖

Quiz: Module quiz: Programming Paradigms
8 questions
- 📖

Reading: Additional resources
5 min

Exercise: Instantiate a custom Object

This is your first experience creating classes and objects in Python. You will be following a sequential process where you will create a class, define its state by creating variables and functions to define its attributes and behavior, and then instantiate it using some variable. Finally, you will use the class members to get the desired output.

Follow the steps to build and run your program in the environment provided at the bottom of the reading.

Step 1

- 1.1 Define a class called **MyFirstClass**.
- 1.2 Add a print statement inside it such as “**Who wrote this?**”.

Step 2

Create a string variable named **index** and initialize it with a string “**Author-Book**”.

Step 3

- 3.1 Define a function called **hand_list()** with the help of **def** keyword.
- 3.2 Pass the parameter **self** to it. And then pass two parameters, **philosopher** and **book** to it.

Step 4

- 4.1 Write a print statement using the **print()** function and pass the class variable by accessing it.
- Hint: Class variables are accessed directly by calling it over the class name using dot notation.**
- 4.2 Write a print statement that will give output such as: “**Plato wrote the book: Republic**” where “Plato” is the philosopher and “Republic” is the book.

Hint: You can make use of the built-in (+) concatenation operator to join these strings.

Step 5

- 5.1 Create and instantiate an object of that class, called **whodunnit**
- 5.2 Call method **hand_list()** over this object “whodunnit” and pass two values to it namely “Sun Tzu” and “The Art of War”.

Use the following code block to build your program:

```
1  # Define class MyFirstClass
2
3      # Define string variable called index
4
5      # Define function hand_list()
6
7          # variable + “ wrote the book: ” + variable
8
9
10 # Call function handlist()
```

Run

Reset

No Output

✓ Completed

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